

D693

Elementary Mathematics Methods and Interventions

Armored Drill

8 concepts · 12 items · 4 transfer items

What this is. Every concept in this course's public description, re-asked with more disguise than a practice test uses. Each stem carries facts that do not decide it. Every wrong answer is a named trap.

Source. Scope from the course's public description in the WGU Institutional Catalog (September 2026); items written from the field's standard body of knowledge. No WGU assessment item, course material or proprietary content is used or reproduced. Not affiliated with or endorsed by Western Governors University.

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How to run it

Paper first. Pencil. One pass, no notes, a timer if you want the pressure.

1. **Before you read the options**, fill the strip under the item: the target word (what is actually being asked), and the one fact in the stem that decides it.
2. **Principle before arithmetic**. Name the rule, then compute.
3. **Camouflage**. Most of every stem does not matter. Ask: which single fact decides this?
4. **Inverse pairs are the bait**. If two options are opposites, one of them is the trap built for you.
5. **Mark sure, unsure or guess** honestly. Sure-and-wrong is the most valuable miss on the page — it is a wrong mental model.
6. **Absolutes** (always, never, guarantee) are usually wrong. And recognizing an answer is not the same as owning it.
7. Items with a **heavy black bar** on the left are **transfer items**: the same principle, reversed or solved for a different variable.
8. Stop at the STOP page. Then grade, fill the triage grid, and tally the traps you fell for. The biggest tally is what you drill next.

The strip, filled in, looks like this:

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	_____	

Items

1.

A fourth grader who is on the swim team says that 0.35 is greater than 0.5 “because 35 is bigger than 5.” The class has been using new decimal grids. What is the best teaching response?

- A.** Tell her the number with more digits is always greater.
- B.** Tell her to compare the last digits.
- C.** Mark it wrong without explanation.
- D.** Have her shade 0.5 and 0.35 on hundredths grids and compare by place value, showing $0.5 = 0.50$.

Target word _____	The one fact that decides _____	Answer _____	☐ sure ☐ unsure ☐ guess
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2. [TRANSFER — reversed stem / different variable]

Which decimal is greatest?

- A.** 0.7
- B.** 0.65
- C.** 0.692
- D.** 0.09

Target word _____	The one fact that decides _____	Answer _____	☐ sure ☐ unsure ☐ guess
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3.

A fifth grader with neat handwriting multiplies $12 \times \frac{1}{2}$ and gets 6, then erases it, saying, “That can’t be right — multiplication makes numbers bigger.” The class sits in table groups of four. How should the teacher address this?

- A.** Show $12 \times \frac{1}{2}$ as “half of 12” with a model, and discuss why multiplying by a number less than 1 gives a smaller product.
- B.** Agree; $12 \times \frac{1}{2}$ should be 24.
- C.** Say multiplication always makes bigger except with fractions, so just memorize it.
- D.** Regroup the tables so she sits with a stronger student.

Target word _____	The one fact that decides _____	Answer _____	☐ sure ☐ unsure ☐ guess
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4. [TRANSFER — reversed stem / different variable]

Without computing, which product is less than 48?

- A.** 48×2
- B.** $48 \times 3/4$
- C.** 48×1
- D.** $48 \times 5/4$

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

5.

A fourth-grade class with a new document camera uses an area model for 23×14 . The school's math night drew 150 families. What is the product, found by partial products?

- A.** 212
- B.** 322
- C.** 150
- D.** 37

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

6. [TRANSFER — reversed stem / different variable]

A student sets up an area model for 36×25 . What are the four partial products?

- A.** 6, 15, 12, 30
- B.** 600 and 30 only
- C.** 600, 150, 120, 30
- D.** 900

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

7.

A third grader at a school with a popular pizza day says $1/4$ can be more than $1/2$ because "my $1/4$ of the large pizza was bigger than my sister's $1/2$ of a personal pizza." The large pizza had pepperoni. What is the key idea to teach?

- A.** The student is right; $1/4$ is sometimes greater than $1/2$ as numbers.
- B.** Pepperoni adds to the size of the slice.
- C.** Fractions can be compared only when they refer to the same-size whole; with equal wholes, $1/2 > 1/4$.
- D.** Fractions with bigger denominators are bigger.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

8.

A fifth-grade class baking for a school fundraiser, which raised \$600 last year, has 3 cups of flour. Each batch uses $\frac{1}{4}$ cup. How many batches can they make?

- A. 12
- B. $\frac{3}{4}$
- C. 600
- D. 4

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

9. [TRANSFER — reversed stem / different variable]

A recipe serving is $\frac{2}{3}$ cup. How many servings are in 4 cups?

- A. 2
- B. 12
- C. $\frac{8}{3}$
- D. 6

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

10.

A sixth grader in a class with a new math word wall solves a hard problem after trying three different strategies over 20 minutes. His parents are both accountants. Which teacher response best supports a growth mindset?

- A. “Good,” with no further comment.
- B. “Of course — math runs in your family.”
- C. “You kept trying different strategies until one worked — that persistence paid off.”
- D. “You’re a natural math genius!”

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

11.

A fifth-grade teacher whose class includes students well below and well above grade level is planning a lesson on area. The classroom has 28 desks and a new projector. Which task best serves all learners at once?

- A.** Assign one closed problem with a single answer to all 28 students.
- B.** “Design as many rectangles as you can with an area of 24 square units. What do you notice about their perimeters?”
- C.** Give struggling students a worksheet of 10 easy problems and advanced students 30 harder ones.
- D.** Have advanced students tutor others all period instead of doing math.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

12.

During a five-minute number talk in a class that just returned from music, students mentally solve $49 + 26$. Several give 75 using different methods. What should the teacher do next?

- A.** Assign 20 more addition problems as a worksheet.
- B.** Show the standard algorithm as the one correct way.
- C.** Praise whoever answered first and move on.
- D.** Ask several students to explain their strategies, record each one, and have the class compare them.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

STOP

Answers and dissection follow. Finish every item and every strip first.

Key and dissection

Struck-through text is the camouflage: facts that do not decide the item.

1. Answer D Concept 1

A fourth grader ~~who is on the swim team~~ says that 0.35 is greater than 0.5 “because 35 is bigger than 5.” ~~The class has been using new decimal grids.~~ What is the best teaching response?

Target word: respond to the decimal misconception **Decides:** $0.5 = 5 \text{ tenths} = 50 \text{ hundredths} > 35 \text{ hundredths}$.

A	Absolute language	Confirms the misconception.
B	Inverted operation	Compare from the largest place (tenths) first, not the last digit.
C	Dropped element	Drops any instruction to fix the thinking.
D	Correct	Visual models and place value expose the “longer is larger” error.

2. Answer A Concept 1 • transfer

Which decimal is greatest?

Target word: order decimals **Decides:** Compare tenths first.

A	Correct	7 tenths beats 6 tenths and 0 tenths.
B	Adjacent category	6 tenths — less than 7 tenths.
C	Confusion set	Most digits, but only 6 tenths.
D	Inverted operation	Largest last digit, smallest value.

3. Answer A Concept 2

A fifth grader ~~with neat handwriting~~ multiplies $12 \times 1/2$ and gets 6, then erases it, saying, “That can’t be right — multiplication makes numbers bigger.” ~~The class sits in table groups of four.~~ How should the teacher address this?

Target word: address the “multiplication makes bigger” idea **Decides:** Multiplying by a fraction less than 1 takes part of the number.

A	Correct	Connects the operation to meaning and generalizes the rule.
B	Inverted operation	That is $12 \div 1/2$.
C	Dropped element	Drops the reason; a rule without meaning will not transfer.
D	Camouflage	Uses the seating detail; the misconception itself is not addressed.

4. Answer B Concept 2 • transfer

Without computing, which product is less than 48?

Target word: predict product size **Decides:** The factor less than 1 shrinks the product.

A	Adjacent category	Doubles the number.
B	Correct	$3/4 < 1$, so the product is less than 48 (it is 36).
C	Confusion set	Equals 48 — not less.
D	Inverted operation	$5/4 > 1$ makes a larger product.

5. Answer B Concept 3

A fourth-grade class ~~with a new document camera~~ uses an area model for 23×14 . ~~The school's math night drew 150 families.~~ What is the product, found by partial products?

Target word: compute with partial products **Decides:** $20 \times 10 + 20 \times 4 + 3 \times 10 + 3 \times 4$.

A	Dropped element	Only tens $\times$ tens and ones $\times$ ones ($200 + 12$); drops the cross products.
B	Correct	$200 + 80 + 30 + 12 = 322$.
C	Wrong stem number	The math-night attendance.
D	Inverted operation	Adds the factors.

6. Answer C Concept 3 • transfer

A student sets up an area model for 36×25 . What are the four partial products?

Target word: list partial products **Decides:** 30×20 , 30×5 , 6×20 , 6×5 .

A	Dropped element	Drops place value (treats 30 as 3 and 20 as 2).
B	Adjacent category	Skips the two cross products.
C	Correct	Sum is $900 = 36 \times 25$.
D	Confusion set	The final product, not the partial products.

7. Answer C Concept 4

A third grader ~~at a school with a popular pizza day~~ says $1/4$ can be more than $1/2$ because “my $1/4$ of the large pizza was bigger than my sister’s $1/2$ of a personal pizza.” ~~The large pizza had pepperoni.~~ What is the key idea to teach?

Target word: the key fraction idea **Decides:** Fractions compare only with the same-size whole.

A	Inverted operation	As numbers, $1/2$ is always greater than $1/4$.
B	Camouflage	Topping is irrelevant.
C	Correct	The wholes differed, so the comparison was not fair.
D	Confusion set	More parts means smaller parts.

8. Answer A Concept 5

A fifth-grade class baking for a school fundraiser, ~~which raised \$600 last year,~~ has 3 cups of flour. Each batch uses $1/4$ cup. How many batches can they make?

Target word: measurement division **Decides:** How many $1/4$ s fit in 3.

A	Correct	$3 \div 1/4 = 3 \times 4 = 12$.
B	Inverted operation	Multiplies $3 \times 1/4$ instead of dividing.
C	Camouflage	The fundraiser amount.
D	Wrong stem number	The denominator, not the number of batches.

9. Answer D Concept 5 · transfer

A recipe serving is $\frac{2}{3}$ cup. How many servings are in 4 cups?

Target word: divide by a non-unit fraction **Decides:** $4 \div \frac{2}{3} = 4 \times \frac{3}{2}$.

A	Dropped element	Divides by the numerator only.
B	Confusion set	Multiplies by the denominator and forgets to divide by 2.
C	Inverted operation	Multiplies by $\frac{2}{3}$ instead of dividing.
D	Correct	$4 \times \frac{3}{2} = 6$.

10. Answer C Concept 6

A sixth grader ~~in a class with a new math word wall~~ solves a hard problem after trying three different strategies over 20 minutes. ~~His parents are both accountants.~~ Which teacher response best supports a growth mindset?

Target word: feedback wording **Decides:** Praise the process — persistence and strategies.

A	Dropped element	Drops specific feedback on the process.
B	Camouflage	Keys on the parents; implies ability is inherited.
C	Correct	Names effort and strategy, which students can repeat.
D	Adjacent category	Praises fixed ability, which can make students avoid challenge.

11. Answer B Concept 7

A fifth-grade teacher whose class includes students well below and well above grade level is planning a lesson on area. ~~The classroom has 28 desks and a new projector.~~ Which task best serves all learners at once?

Target word: task for mixed readiness **Decides:** Low floor, high ceiling.

A	Wrong stem number	Uses the desk count; a single closed item has no high ceiling.
B	Correct	Everyone can start with one rectangle; the pattern extends to deep reasoning.
C	Adjacent category	Differentiates quantity, not access or depth.
D	Inverted operation	Removes challenge for advanced learners.

12. Answer D Concept 8

During a five-minute number talk ~~in a class that just returned from music~~, students mentally solve $49 + 26$. Several give 75 using different methods. What should the teacher do next?

Target word: the next number-talk move **Decides:** Surface, record, and compare strategies.

A	Adjacent category	Practice, not discourse.
B	Absolute language	Number talks value multiple valid strategies.
C	Dropped element	Drops the sharing of strategies; rewards speed.
D	Correct	Justifying and comparing strategies is the heart of a number talk.

Triage grid

Write each item number in the box it belongs to.

	Right	Wrong
Sure	Owled.	Concept inversion — top priority. A wrong mental model; it survives rewording.
Unsure / guess	Recognition, not mastery. Re-drill with the transfer.	Not learned yet.

Trap tally

For every miss, find the letter you chose in the key and tally its trap. The biggest number is the pattern to drill.

Trap	What it does	In this drill	Your misses
Dropped element	Leaves out a required term or step.	7	
Wrong stem number	A figure copied from the prompt that is not the answer.	3	
Adjacent category	The right idea in the wrong bucket.	6	
Inverted operation	Direction flipped: add/subtract, cause/effect, higher/lower.	9	
Absolute language	Always, never, guarantee — where judgment applies.	2	
Confusion set	Every option is real; the test is exact role-matching.	5	
Camouflage	Answers a stem fact that was never the question.	4	

Anchor rows — the concept list

One row per concept: the rule to lock in, the trap that catches people, and the tell in the question that signals it.

#	Concept	Rule	Trap	Tell
1	“Longer is larger” decimal misconception	Compare decimals by place value (tenths first), not by the number of digits: $0.5 > 0.35$.	Treating the digits after the point like a whole number.	Decimals with different numbers of digits.
2	Multiplying by a fraction less than 1	Multiplying a positive number by a fraction less than 1 gives a smaller product; by a fraction greater than 1, a larger one.	“Multiplication always makes bigger.”	A student surprised that $12 \times \frac{1}{2}$ is less than 12.
3	Area model and partial products	Break factors into place-value parts, multiply each pair, and add: $23 \times 14 = 200 + 30 + 80 + 12$.	Multiplying only tens by tens and ones by ones.	A two-digit by two-digit multiplication shown in a grid.
4	Fractions need the same-size whole	Fractions can only be compared or combined when they refer to the same-size whole; equal parts must be equal in size.	Comparing $\frac{1}{2}$ of a small pizza with $\frac{1}{4}$ of a large one as pure fractions.	Two fraction models with different wholes, or unequal parts.
5	Dividing by a fraction	$a \div b/c$ asks how many b/c -size groups fit in a (measurement division); it equals $a \times c/b$.	Multiplying by the fraction instead of dividing, or dividing only by the numerator.	“How many servings / pieces of size ___ fit?”
6	Growth-mindset feedback	Praise effort, strategies, and persistence (process), and treat mistakes as information for learning.	Praising intelligence (“You’re so smart”).	A teacher’s words after a student succeeds or struggles.
7	Low-floor, high-ceiling tasks	Open tasks that everyone can start and that extend to deep challenge let one task serve diverse learners.	Separate, easier worksheets for struggling students and more problems for advanced ones.	A teacher planning for a wide range of readiness in one lesson.
8	Number talks and meaningful discourse	In a number talk, students solve mentally, share and justify strategies, and the teacher records and compares them without judging speed.	Calling on the fastest student and moving on.	A brief mental-math routine with several strategies.

Course scope (public description): Elementary Mathematics Methods and Interventions is an engaging course crafted for educators who are dedicated to strengthening and enriching the mathematics education of elementary students. This course focuses on practical, research-supported strategies for teaching essential mathematical concepts, including number operations, geometry, measurement, and data analysis. This course includes resources for all elementary grade levels with a significant focus on grades 3 to 6. Participants learn to identify and assess students' mathematical understanding, create differentiated learning experiences, and implement effective interventions for diverse learner needs. Emphasis is placed on constructing a classroom environment that promotes mathematical curiosity, problem-solving, meaningful discourse, and a growth mindset. Students acquire the skills to build authentic mathematics experiences that caters to the varied needs of all students, setting a firm foundation for their future academic success in mathematics.